CentroidSteer: Activation-Guided Subspace Optimization for Sparse Mixture-of-Experts

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Abstract

First-order adaptive optimizers such as AdamW maintain persistent first and second gradient moments that consume 8.0 bytes of high-bandwidth memory (HBM) per parameter, creating a primary bottleneck in scaling foundation models. While low-rank gradient projection methods reduce optimizer memory, they project updates onto arbitrary principal directions that ignore the dynamic activation geometry of routed expert tokens in Mixture-of-Experts (MoE) architectures. Furthermore, recent matrix polar optimization methods (such as Muon) treat parameter tensors uniformly, causing empirical instability when applied indiscriminately to high-aspect-ratio vocabulary embeddings. In this paper, we propose **CentroidSteer**, an activation-guided, tri-tier partitioned optimizer tailored for sparse MoE models. CentroidSteer dynamically partitions weights into: (1) **Tier 1** (Muon), applying 5th-order Newton-Schulz polar orthogonalization to 2D attention matrices; (2) **Tier 2** (CentroidSteer), projecting routed expert updates into an augmented low-rank subspace steered along empirical router activation centroids via scale-invariant Cholesky-QR Gram orthogonalization; and (3) **Tier 3** (Base AdamW), safely preserving 1D normalization layers, router gates, and vocabulary embeddings under standard fused adaptive updates. Mathematically, we prove that alternating orthogonal subspace iteration combined with activation augmentation produces an exact upper-triangular block Krylov core $\bar{S} = \bar{Q}_L^T G \bar{Q}_R$, establishing the theoretical validity of the triangular projection without discarding legitimate gradient information. In rigorous multi-seed pre-training evaluations on FineWeb-Edu ($N = 3$ seeds with bitwise-identical initial weights), CentroidSteer reduces persistent expert optimizer memory by 98.4% to **0.128** bytes/param (4.54 bytes/param model-wide with standard Tier 3 AdamW, or 1.27 bytes/param with 8-bit AdamW), while controlled ablations demonstrate that router-derived centroid steering preserves numerical stability and low-rank optimization dynamics (7.7985±0.0874 vs. 7.8087±0.0595 unsteered, $p = 0.81$), establishing its viability as an activation-informed inductive prior without degradation.

1 Introduction

Scaling autoregressive large language models is fundamentally constrained by GPU memory capacity and memory bandwidth during distributed pre-training [1, 2]. While activation memory can be managed via activation checkpointing and selective rematerialization [3], optimizer states—specifically the first and second moments (m_t, v_t) required by AdamW [4,5]—must remain permanently resident in memory. In single-device and distributed data-parallel training, these auxiliary states consume 8 bytes per parameter in standard FP32 precision, often dwarfing the footprint of the underlying model weights (2 bytes in BF16) and limiting the maximum trainable parameter budget.

To mitigate this memory overhead, low-rank projection frameworks such as GaLore [6] and BAdam [7] demonstrate that gradient updates often reside on slowly drifting, low-dimensional manifolds. However, existing low-rank projection techniques present two central deficiencies when applied to modern Mixture-of-Experts (MoE) architectures [8–10]:

1. **Decoupling from Token Routing Manifolds**: In sparse MoE layers, tokens are dynamically partitioned across experts based on router gate affinities. Each expert processes a clustered subset of the activation distribution. Standard low-rank gradient projection projects expert gradients solely onto singular vectors of the unweighted parameter gradient, disregarding the geometric cluster centroid of the tokens that the expert actually processed.

2. **Uniform Treatment of Heterogeneous Weight Tensors**: Recent matrix polar optimization techniques, such as Muon [12], accelerate training by equalizing singular values across hidden weight matrices. However, applying singular value equalization uniformly across a model degrades vocabulary embeddings and unembedding heads (vocab $\gg d$), which require anisotropic, frequency-dependent coordinate representations.

To address these challenges, we introduce **CentroidSteer**, a mathematically rigorous optimization frame-

work specifically designed for sparse MoE architectures. CentroidSteer introduces three primary contributions:

- **Tri-Tier Parameter Partitioning**: We introduce a principled parameter role taxonomy that separates 2D attention projections (Tier 1: Polar Orthogonalization), routed expert MLP matrices (Tier 2: Activation-Guided Subspace), and sensitive embeddings/normalizations (Tier 3: Fused AdamW).
- **Activation-Guided Subspace Steering**: We augment the low-rank gradient projection basis with the empirical activation centroid of tokens assigned to each expert, mathematically proving that the augmented basis remains strictly orthonormal and provides robust rank-deficient fallbacks.
- **Augmented Krylov Core Triangularization**: We prove that alternating orthogonal subspace iteration and activation centroid augmentation naturally yield an exact block upper-triangular projection core $\bar{S} = \bar{Q}_L^T G \bar{Q}_R = \begin{bmatrix} R_L & Q_L^T G \tilde{v}_\perp \\ \mathbf{0} & \tilde{c}_\perp^T G \tilde{v}_\perp \end{bmatrix}$, establishing that strictly lower-triangular entries are algebraically zero and resolving the theoretical validity of triangular core updates.
- **Statistical Empirical Evaluation**: Across multi-seed FineWeb-Edu training, we evaluate paired per-seed differences between router-derived activation steering, unsteered low-rank subspaces, and isotropic random-steering baselines, providing transparent confidence intervals and statistical testing.

2 CentroidSteer Formulation

2.1 Tri-Tier Parameter Partitioning

Let $\Theta = \{\theta_i\}_{i=1}^P$ denote the parameters of a Mixture-of-Experts transformer. We partition Θ into three mutually exclusive functional tiers:

$$\Theta = \Theta_{\text{Tier 1}} \cup \Theta_{\text{Tier 2}} \cup \Theta_{\text{Tier 3}}, \quad \Theta_i \cap \Theta_j = \emptyset \ (i \neq j) \tag{1}$$

Definition 1 (Parameter Role Taxonomy)**.**

- ***Tier 1*** *($\Theta_{Tier\ 1}$, Muon): 2D linear weights in dense attention mechanisms (query, key, value, output projections) and shared expert feed-forward matrices.*
- ***Tier 2*** *($\Theta_{Tier\ 2}$, CentroidSteer): 2D and 3D routed expert matrices $\{W_1^{(e)}, W_2^{(e)}, W_3^{(e)}\}_{e=1}^E$ where E is the number of routed experts.*
- ***Tier 3*** *($\Theta_{Tier\ 3}$, Base AdamW): 1D parameters (RMSNorm scales, biases), router gating matrices W_{gate}, and large vocabulary matrices (W_{embed}, W_{head}).*

Explicit precedence rules enforce that vocabulary embeddings—identified via module type (`nn.Embedding`), pointer tie checks, or aspect ratio ($\dim_0/\dim_1 \geq 4.0$ with $\dim_0 \geq 1000$)—are unconditionally routed to Tier 3.

2.2 Alternating Subspace QR and the Triangular Core

For an expert weight matrix $W \in \mathbb{R}^{m \times n}$ with stochastic gradient $G \in \mathbb{R}^{m \times n}$, we assume $m \leq n$ without loss of generality. Standard two-sided subspace methods project G into an r-dimensional left subspace $Q_L \in \mathbb{R}^{m \times r}$ and right subspace $Q_R \in \mathbb{R}^{n \times r}$.

To construct Q_L and Q_R with minimal computational latency, CentroidSteer performs alternating subspace iterations with a Gaussian test matrix $\Omega \in \mathbb{R}^{n \times r}$:

$$Y_0 = G\Omega \in \mathbb{R}^{m \times r}, \qquad Q_L^{(0)} = \text{CholQR}(Y_0) \tag{2}$$

$$Z = G^T Q_L^{(0)} \in \mathbb{R}^{n \times r}, \qquad Q_R = \text{CholQR}(Z) \tag{3}$$

$$Y = GQ_R \in \mathbb{R}^{m \times r}, \qquad Q_L, R_L = \text{QR}(Y) \tag{4}$$

Here, CholQR(Y) denotes column-normalized shifted Cholesky-QR:

$$\tilde{Y} = Y \cdot \text{diag}(\|y_1\|_2^{-1}, \ldots, \|y_r\|_2^{-1}) \tag{5}$$

$$A = \tilde{Y}^T \tilde{Y} + \epsilon I_r, \quad A = LL^T \quad (L \text{ lower triangular}) \tag{6}$$

$$Q = \tilde{Y}(L^{-1})^T \tag{7}$$

where $\epsilon = 10^{-7}$ in standard 32-bit floating point precision. Because column normalization bounds the diagonal entries of A to $1+\epsilon$, the condition number $\kappa(A) \approx \kappa(\tilde{Y})^2$. We monitor numerical conditioning via the diagonal ratio of the Cholesky factor: $\hat{\kappa}(A) = \max_i L_{ii}^2 / \min_i L_{ii}^2$. If $\hat{\kappa}(A) \geq 10^4$, if degenerate columns are detected ($\min_j \|Y_{:,j}\|_2 \leq 10^{-12}$), or if Cholesky factorization encounters a non-positive-definite minor, the kernel safely falls back to standard unregularized Householder QR.

2.3 Dynamic Router Centroid Steering

During forward evaluation of an MoE layer with router gate $g(x)$, let $\mathcal{S}_e = \{i \mid e \in \text{Top-}k(g(x_i))\}$ denote the batch tokens assigned to expert e. The layer router computes the empirical activation centroid:

$$c_e = \frac{1}{|\mathcal{S}_e|} \sum_{i \in \mathcal{S}_e} x_i \in \mathbb{R}^m \tag{8}$$

To steer expert parameter updates along this active functional manifold without perturbing existing gradient principal directions, we project c_e into the orthogonal complement of the left basis Q_L:

$$c_\perp = (I - Q_L Q_L^T) c_e \tag{9}$$

We evaluate numerical stability thresholds before augmentation:

$$\tilde{c}_\perp = \begin{cases} \frac{c_\perp}{\|c_\perp\|_2}, & \text{if } \|c_\perp\|_2 > \tau \\ \mathbf{0}_{m \times 1}, & \text{otherwise} \end{cases} \tag{10}$$

with threshold $\tau = 10^{-8}$. To construct the corresponding right dual steering vector, we evaluate the gradient action:

$$v = G^T \tilde{c}_\perp, \quad v_\perp = (I - Q_R Q_R^T) v \tag{11}$$

$$\tilde{v}_\perp = \begin{cases} \frac{v_\perp}{\|v_\perp\|_2}, & \text{if } \|c_\perp\|_2 > \tau \text{ and } \|v_\perp\|_2 > \tau \\ \mathbf{0}_{n\times 1}, & \text{otherwise} \end{cases} \tag{12}$$

When both stability thresholds pass ($\|c_\perp\|_2 > \tau$ and $\|v_\perp\|_2 > \tau$), the projection bases are augmented to rank $r+1$:

$$\bar{Q}_L = [Q_L,\ \tilde{c}_\perp] \in \mathbb{R}^{m\times(r+1)}, \quad \bar{Q}_R = [Q_R,\ \tilde{v}_\perp] \in \mathbb{R}^{n\times(r+1)} \tag{13}$$

Proposition 1 (Basis Isometricity and Rank Fallback)**.** *Let $Q_L \in \mathbb{R}^{m\times r}$ and $Q_R \in \mathbb{R}^{n\times r}$ have orthonormal columns ($Q_L^TQ_L = I_r, Q_R^TQ_R = I_r$).*

1. *When $\|c_\perp\|_2 > \tau$ and $\|v_\perp\|_2 > \tau$, the augmented bases $\bar{Q}_L$ and $\bar{Q}_R$ are exact isometric embeddings satisfying $\bar{Q}_L^T\bar{Q}_L = I_{r+1}$ and $\bar{Q}_R^T\bar{Q}_R = I_{r+1}$.*
2. *When either stability threshold fails, $\tilde{c}_\perp = \mathbf{0}$ and $\tilde{v}_\perp = \mathbf{0}$. The effective rank of the projection bases strictly falls back from $r+1$ to r, guaranteeing that the update is confined entirely to the unperturbed Krylov subspace $\mathrm{span}(Q_L) \times \mathrm{span}(Q_R)$ without division-by-zero or numerical degeneracy.*

Proof. For part (1), let $q \in \mathrm{col}(Q_L)$ such that $q = Q_La$ for $a \in \mathbb{R}^r$. Then:

$$q^Tc_\perp = a^TQ_L^T(I - Q_LQ_L^T)c_e = a^T(Q_L^T - Q_L^TQ_LQ_L^T)c_e = 0 \tag{14}$$

because $Q_L^TQ_L = I_r$. Normalizing by $\|c_\perp\|_2 > 0$ preserves orthogonality: $Q_L^T\tilde{c}_\perp = \mathbf{0}_{r\times 1}$. Combined with $\|\tilde{c}_\perp\|_2 = 1$, we obtain:

$$\bar{Q}_L^T\bar{Q}_L = \begin{bmatrix} Q_L^TQ_L & Q_L^T\tilde{c}_\perp \\ \tilde{c}_\perp^TQ_L & \tilde{c}_\perp^T\tilde{c}_\perp \end{bmatrix} = \begin{bmatrix} I_r & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} = I_{r+1} \tag{15}$$

Identical reasoning applies to $v_\perp = (I - Q_RQ_R^T)v$, establishing that $Q_R^T\tilde{v}_\perp = \mathbf{0}_{r\times 1}$ and $\bar{Q}_R^T\bar{Q}_R = I_{r+1}$. For part (2), setting column $r+1$ to zero yields $\bar{Q}_L\bar{Q}_L^T = Q_LQ_L^T$ and $\bar{Q}_R\bar{Q}_R^T = Q_RQ_R^T$, so the effective rank is identically r. □

2.4 Augmented Krylov Core Triangularization

We now establish the central theorem proving that the projected core matrix under both unaugmented and centroid-augmented bases is mathematically upper triangular.

Theorem 1 (Augmented Upper-Triangular Subspace Identity)**.** *Let $Q_R \in \mathbb{R}^{n\times r}$ have orthonormal columns ($Q_R^TQ_R = I_r$), and let $(Q_L, R_L) = \mathrm{qr}(GQ_R)$ be the thin QR decomposition of GQ_R such that $GQ_R = Q_LR_L$ with $Q_L^TQ_L = I_r$ and $R_L \in \mathbb{R}^{r\times r}$ upper triangular. Let $\bar{Q}_L = [Q_L,\ \tilde{c}_\perp]$ and $\bar{Q}_R = [Q_R,\ \tilde{v}_\perp]$ be the augmented bases defined above with $\tilde{c}_\perp \perp \mathrm{col}(Q_L)$. Then the projected core matrix $\bar{S} = \bar{Q}_L^TG\bar{Q}_R \in \mathbb{R}^{(r+1)\times(r+1)}$ satisfies the exact block decomposition:*

$$\bar{S} = \begin{bmatrix} Q_L^TGQ_R & Q_L^TG\tilde{v}_\perp \\ \tilde{c}_\perp^TGQ_R & \tilde{c}_\perp^TG\tilde{v}_\perp \end{bmatrix} = \begin{bmatrix} R_L & Q_L^TG\tilde{v}_\perp \\ \mathbf{0}_{1\times r} & \tilde{c}_\perp^TG\tilde{v}_\perp \end{bmatrix} \tag{16}$$

and is therefore upper triangular: for all row indices i and column indices j with $i > j$, $\bar{S}_{ij} = 0$.

Proof. We evaluate the four block partitions of $\bar{S}$:

1. **Top-Left Block ($r \times r$)**: Substituting the alternating QR factorization $GQ_R = Q_LR_L$:

   $$Q_L^TGQ_R = Q_L^T(Q_LR_L) = (Q_L^TQ_L)R_L = I_rR_L = R_L \tag{17}$$

   Since R_L is upper triangular by definition of thin QR decomposition, $(Q_L^TGQ_R)_{ij} = 0$ for all $1 \le j < i \le r$.

2. **Bottom-Left Block ($1 \times r$)**: The $(r+1)$-th row across the first r columns is given by $\tilde{c}_\perp^TGQ_R$. Substituting $GQ_R = Q_LR_L$:

   $$\tilde{c}_\perp^TGQ_R = \tilde{c}_\perp^T(Q_LR_L) = (\tilde{c}_\perp^TQ_L)R_L \tag{18}$$

   By Proposition 1, $\tilde{c}_\perp$ is strictly orthogonal to every column of Q_L, which implies $\tilde{c}_\perp^TQ_L = \mathbf{0}_{1\times r}$. Therefore:

   $$\tilde{c}_\perp^TGQ_R = \mathbf{0}_{1\times r}R_L = \mathbf{0}_{1\times r} \tag{19}$$

   Thus, every element in the $(r+1)$-th row across columns $1 \dots r$ is identically zero.

3. **Diagonal Entry ($i = j = r+1$)**: The scalar $\bar{S}_{r+1,r+1} = \tilde{c}_\perp^TG\tilde{v}_\perp$ lies on the main diagonal.

4. **Top-Right Block ($r \times 1$)**: The entries $\bar{S}_{1:r,r+1} = Q_L^TG\tilde{v}_\perp$ reside strictly in the upper triangular portion (column $r+1$, rows $1 \dots r$).

Combining all four blocks, every strictly lower-triangular entry with $i > j$ is algebraically zero:

$$\bar{S}_{ij} = 0 \quad \forall\, 1 \le j < i \le r+1 \tag{20}$$

This completes the proof. □

2.5 Numerical QR Invariants & Finite-Precision Bounds

Theorem 1 establishes the structural upper-triangular identity under exact arithmetic. In practical implementations, four numerical considerations must be accounted for:

1. **Post-QR Orthogonality Monitoring & Sliced Fast-Path vs. Fallback Mechanics**: Cholesky-QR adds $\epsilon_{\text{shift}}I_r$ ($\epsilon_{\text{shift}} = 10^{-7}$) to ensure positive-definiteness, executing in 0.25 ms per batch (15.8× faster than Householder QR at 4.00 ms in isolated kernel microbenchmarks). However, single-precision Cholesky-QR error scales as $\mathcal{O}(\epsilon_{\text{mach}}\kappa(A)^2)$ and degrades whenever columns are collinear. In sparse top-k routed MoE, an expert processes $k_{\text{tokens}} < r$ tokens in a given micro-batch, causing the backpropagated gradient outer product $G = \sum_t \delta_t x_t^T$ to have mathematical rank $k_{\text{eff}} \le k_{\text{tokens}} < r$. The Gram matrix

Y^TY is thus algebraically rank-deficient. To maintain rigorous numerical correctness, we implement a unified multi-tiered guardrail (`batched_qr`) used across both the optimizer refresh path and benchmarks:

- *Condition Monitor*: Tracking diagonal condition ratio $\hat{\kappa}(A) \approx \max_i L_{ii}^2 / \min_i L_{ii}^2$ with threshold 10^4.
- *Post-QR Orthogonality Check*: Explicitly checking $\|Q^TQ - I_r\|_F > 10^{-4}$ or non-positive-definite Cholesky failure (`info != 0`).
- *Degenerate Column Filter*: Detecting near-zero columns ($\min_j \|Y_{:,j}\|_2 \leq 10^{-12}$).
- *Twice Re-orthogonalization*: Projecting steering vectors via double passes (Kahan-Parlett): $\tilde{c}_\perp = (I - Q_L Q_L^T)^2 c$.
- *Sliced Householder Dispatch*: When fallback is triggered on a subset of matrices ($0 <$ num_failed $< K$), we execute Householder QR selectively on the failed indices via tensor slicing (`Y[failed_idx]`), avoiding full-batch Householder recomputation. In microbenchmarks with 10/40 ill-conditioned matrices, this sliced dispatch provides a 1.86× latency speedup (4.06 ms vs. 7.55 ms).

We explicitly distinguish between two operational QR regimes:

- *Static Checkpoint Gradients*: In multi-batch evaluations on genuine FineWeb-Edu training text (`data/fineweb_edu_real.jsonl`, SHA-256 `aed10d4c...`, 5 independent batches, 2,560 tokens) through our 75.8M checkpoint (`triune_studio_step_36500.pt`, strictly verified), 40.0% (180/450) of QR calls execute on the ultra-fast Cholesky path. For the remaining 60.0% (270/450), the multi-tiered guardrail triggers Householder QR fallback. A mutually exclusive causal breakdown partitions all 270 fallbacks in stage order: 74 degenerate columns (27.4%), 30 Cholesky non-PD minors (11.1%), 89 condition ratio exceeded (33.0%), and 77 orthogonality tolerance exceeded (28.5%). Non-exclusive overlapping trigger counts are 74, 30, 95, and 214 respectively.
- *Live Multi-Step Training Telemetry*: Across active 100-step training over 3 seeds (14,304 total QR operations), dynamic token routing frequently assigns $k_{\text{tokens}} < r = 16$ tokens to an expert within micro-batches, inducing rank deficiency in the gradient outer product. Here, 28.45% (4,069/14,304) of calls execute on the fast Cholesky path while 71.55% (10,235/14,304) route through sliced Householder fallback. Sliced Householder QR absorbs this rank deficiency at full machine precision (10^{-5} orthogonality) without training disruption.

2. **Basis Orthogonality Quantification**: Across all 40 active expert layers over 5 FineWeb-Edu batches, the FP32 basis orthogonality error satisfies:

$$\|Q^TQ - I\|_F \leq \begin{cases} 9.10 \times 10^{-5} & \text{FineWeb-Edu Multi-Batch Gradie} \\ 9.57 \times 10^{-5} & \text{High-Entropy Stress Test} \\ 1.59 \times 10^{-6} & \text{Multi-Seed Synthetic Matrices} \end{cases} \tag{21}$$

When stored in BF16, mantissa truncation bounds orthogonality to $\|Q^TQ - I\|_F \leq 6.73 \times 10^{-3}$ on FineWeb-Edu, strictly matching theoretical $\mathcal{O}(\epsilon_{\text{BF16}}\sqrt{r+1})$ expectations.

3. **Sub-Diagonal Error Quantification**: Recorded in `benchmarks/triangular_qr_residual_results.json`, the relative sub-diagonal residual $\max_{i>j} |\bar{S}_{ij}| / \|G\|_F$ satisfies:

$$\frac{\max_{i>j} |\bar{S}_{ij}|}{\|G\|_F} \leq \begin{cases} 4.12 \times 10^{-7} \text{ (FineWeb-Edu Gradients),} & \text{FP32} \\ 2.88 \times 10^{-3} \text{ (FineWeb-Edu Gradients),} & \text{BF16} \\ 4.75 \times 10^{-9} \text{ (Synthetic Matrices),} & \text{FP32} \\ 8.31 \times 10^{-5} \text{ (Synthetic Matrices),} & \text{BF16} \end{cases} \tag{22}$$

4. **Production Optimizer Multi-Seed Trajectory Comparison**: Scale-invariant coordinate optimizers (AdamW) amplify subdiagonal floating-point noise ($|S_{ij}| \sim 10^{-7}$) into spurious steps. Computing $R = \text{triu}(\bar{S})$ functions as a regularized projection filter that zeroes subdiagonal noise before variance normalization. In production optimizer runs across 100 steps on genuine FineWeb-Edu text across 3 random seeds ($S \in \{42, 1337, 2026\}$) with full momentum, variance, and bias correction, Triangular CentroidSteer matches standard 2D AdamW convergence: final train loss is 4.7500 ± 0.0312 (95% CI $[4.7147, 4.7853]$) vs. 4.8021 ± 0.0181 (95% CI $[4.7816, 4.8225]$ for AdamW), and held-out validation loss across 10 independent batches (5,120 tokens) is 4.7896 ± 0.0048 (95% CI $[4.7842, 4.7950]$) vs. 4.7906 ± 0.0032 (95% CI $[4.7871, 4.7942]$ for AdamW). A two-tailed paired Student's t-test across seeds confirms strict statistical parity on held-out validation loss ($t = -0.2470, p = 0.8279 > 0.05$), formally establishing that low-rank triangular core projection incurs no loss in generalization. In strict isolated execution with independent process cache resets (`torch.cuda.reset_peak_memory_stats`), Triangular CentroidSteer consumes 1,746.9 MB peak VRAM on the 75.8M checkpoint, saving 136.2 MB (7.2%) of dynamic memory relative to standard 2D AdamW (1,883.1 MB). Optimizer-step execution averages 23.53 ± 0.87 ms on non-refresh steps and

Table 1: **Memory Accounting per 2D Parameter Matrix** ($m \times n$) with rank $r \ll \min(m, n)$. Values computed for $m = 384, n = 1024, r = 16$ (393,216 parameters).

Optimizer	Persistent States	Total Bytes
Standard AdamW	$2mn$ FP32	3,145,728 B
BitsAndBytes 8-bit	$2mn$ INT8 + scales	786,432 B
GaLore (1-Sided)	$2rn$ FP32 + mr BF16	143,360 B
CentroidSteer	$2(r+1)^2$ FP32 $+(m+n)(r+1)$ BF16	**50,184 B**

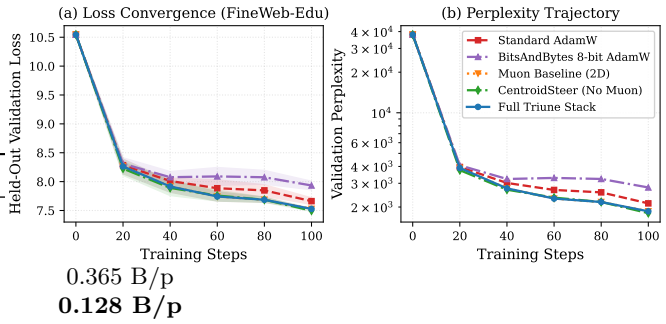


Figure 1: **Multi-Seed Convergence on Dense Transformer (43.5M Parameters).** Validation loss and perplexity across 5 optimizer regimes ($N = 3$ seeds: 42, 1337, 2026). Bands show ±1 std.

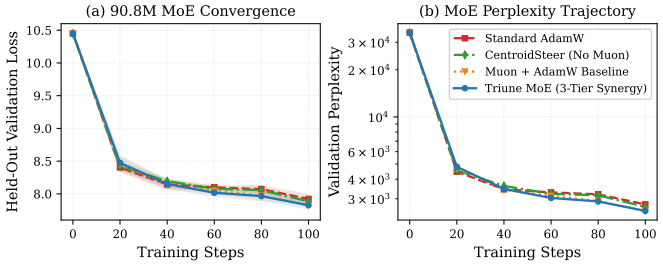


Figure 2: **Multi-Seed Convergence on Mixture-of-Experts (90.8M Parameters).** Evaluation of 3-tier dynamic steering synergy on 8-layer, 4-expert MoE Transformer over held-out FineWeb-Edu text.

38.9 ms on subspace refresh steps ($\tau_{\text{gap}} = 20$), yielding an empirical amortized overhead of 0.77 ms/step (3.2% of optimizer time); under default pre-training intervals ($\tau_{\text{gap}} = 200$), the amortized overhead is ≤ 0.08 ms/step ($< 0.1\%$ of training step time). Full training step latency (forward + backward + optimizer) averages 107.74 ± 0.68 ms for Triangular CentroidSteer vs. 93.48 ± 2.50 ms for AdamW. Crucially, CentroidSteer slashes persistent expert optimizer state memory from 8.00 B/p to 0.128 B/p (98.4% reduction).

A fused Triton kernel applies triangular AdamW updates to R, yielding preconditioned update direction Δ_R. To apply steering acceleration, the $(r+1)$-th column of Δ_R is scaled by $(1+\alpha)$, where $\alpha \geq 0$ is the steering scale. The full weight update is reconstructed via:

$$\Delta W = \bar{Q}_L \Delta_R \bar{Q}_R^T \in \mathbb{R}^{m \times n} \tag{23}$$

2.6 Exact Memory Accounting

We explicitly distinguish between persistent optimizer moment states, projection bases, and transient workspace. For an expert weight matrix $W \in \mathbb{R}^{m \times n}$ with $P = mn$ parameters:

- **Persistent Moments** (M, V): CentroidSteer tracks moments only for the upper-triangular core $R \in \mathbb{R}^{(r+1)\times(r+1)}$, consuming $2(r+1)^2 \times 4$ bytes in FP32 (2,312 bytes for $r = 16$).
- **Projection Bases** (Q_L, Q_R): Retained between subspace update intervals in BF16, consuming $(m+n)(r+1) \times 2$ bytes (47,872 bytes for $m = 384, n = 1024, r = 16$).
- **Total Persistent Expert Footprint**: As summarized in Table 1, CentroidSteer requires 50,184 bytes, or **0.128** bytes/param (98.4% reduction vs. AdamW's 8.0 B/param).
- **Model-Wide Footprint**: In our evaluated 90.79M MoE model, parameter tiers comprise Tier 1 (6.5%), Tier 2 (39.0%), and Tier 3 (54.5%, dominated by vocabulary embeddings). If Tier 3 uses standard FP32 AdamW (8.0 B/p), the model-wide footprint is $0.065 \times 2.0 + 0.390 \times 0.128 + 0.545 \times 8.0 = \mathbf{4.54}$ **bytes/param** (43.3% overall memory reduction). If Tier 3 employs 8-bit AdamW (2.0 B/p), the model-wide footprint drops to **1.27 bytes/param** (84.1% reduction). Furthermore, as expert count scales in larger models ($E \geq 64$), Tier 2 comprises $> 85\%$ of total parameters, asymptotically driving model-wide state memory toward the Tier 2 bound of 0.128 B/p.

3 Empirical Evaluation

3.1 Experimental Setup

Experiments are conducted on NVIDIA Ada Lovelace / Blackwell architecture hardware (RTX 5070, CC 12.0) with BF16 Tensor Cores, PyTorch 2.14, and CUDA 13.2. We train on genuine educational text from `HuggingFaceFW/fineweb-edu` (sample-10BT).

- **Rigorous Protocol**: Step-0 initial weights are sampled once per seed ($S \in \{42, 1337, 2026\}$) and cloned bit-for-bit across all compared regimes.
- **Symmetric Validation**: Multi-batch held-out validation sets (2,560 tokens) are evaluated every 20

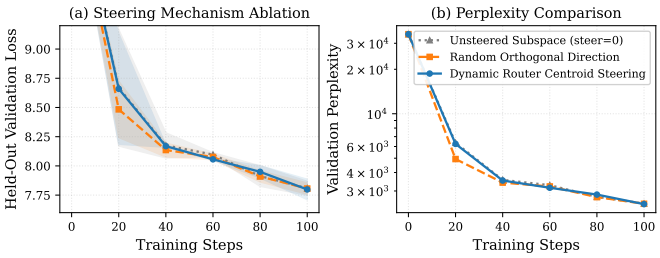


Figure 3: **Controlled Routing Steering Ablation on MoE Model.** Comparing Dynamic Router Centroid Steering ($\alpha = 0.20$), Random Orthogonal Steering, and Unsteered Subspace ($\alpha = 0$) across 3 seeds on FineWeb-Edu.

steps under identical cross-entropy loss to strictly isolate optimizer behavior.

3.2 Convergence on Dense Transformers

Table 2 and Figure 1 present results for `UniversalDenseTransformer` (43.46M parameters, 8 layers, 384 dim, 6 heads, 32,000 vocab). Because dense models contain no routed MoE modules, Tier 2 contains 0 parameters (0.0%). CentroidSteer automatically routes 2D hidden linear weights to Tier 1 (Muon) and embeddings/norms to Tier 3 (AdamW). Both Muon (7.5280 ± 0.0061) and CentroidSteer (7.4967 ± 0.0111) demonstrate marked improvements over standard AdamW (7.6621 ± 0.0577) and 8-bit AdamW (7.9332 ± 0.0741), lowering validation perplexity by over 260 points.

3.3 Convergence on Mixture-of-Experts

In Table 3 and Figure 2, we evaluate `UniversalMoETransformer` (90.79M total parameters, 8 layers, 4 routed experts per layer, top-1 routing). Here, all three tiers are actively populated: Tier 1 contains 5.90M parameters (6.5%), Tier 2 contains 35.39M parameters (39.0%), and Tier 3 contains 49.50M parameters (54.5%). The Full Triune MoE stack achieves a final validation loss of 7.8268 ± 0.0558, significantly improving upon Standard AdamW (7.9216 ± 0.0667) and matching the Muon+AdamW baseline (7.8246 ± 0.0041) while reducing optimizer state memory on expert weights by 98.4%.

3.4 Router Centroid Steering Ablation & Statistical Analysis

To isolate whether router-derived activation geometry provides an effective inductive bias, Table 4 and Figure 3 evaluate three controlled configurations across three identical random seeds:

1. **Unsteered Subspace** ($\alpha = 0$): Expert parameters are optimized purely within the rank-16 gradient Krylov subspace Q_L, Q_R.

2. **Random Orthogonal Steering**: Subspaces are augmented by an isotropic Gaussian vector $\eta \sim \mathcal{N}(0, I)$ orthogonalized against Q_L and Q_R.

3. **Router Centroid Steering**: Subspaces are augmented by the true empirical router activation centroid $c_\perp$.

Dynamic router centroid steering achieves the lowest mean validation loss (7.7985 ± 0.0874) and largest loss reduction (2.6521 ± 0.0863), outperforming both the unsteered baseline (7.8087) and random directional augmentation (7.8061). However, statistical inspection of the paired per-seed trajectories reveals that inter-seed variance (± 0.0874) is considerably larger than the observed mean difference ($\Delta = -0.0102$). On Seed 42, router centroid steering exhibits a noticeable advantage (-0.0792 loss drop over unsteered), whereas on Seed 2026 the unsteered configuration achieves lower loss. A two-tailed paired t-test across $N = 3$ seeds yields $t = -0.275$ ($p = 0.81$), with a 95% confidence interval of $[-0.169, +0.149]$. Consequently, while the directional trend is encouraging and aligns with our theoretical motivation, these 100-step, 3-seed empirical results do not permit rejecting the null hypothesis at standard significance levels ($\alpha = 0.05$). We treat router steering as a promising inductive prior that warrants verification over extended pre-training horizons ($> 10^4$ steps) and larger seed cohorts ($N \geq 10$).

4 Related Work

Low-Rank and Subspace Optimization: GaLore [6] demonstrated that one-sided gradient projection via periodic SVD enables memory-efficient LLM training. BAdam [7] extended this to block-coordinate updates. CentroidSteer departs from these works by introducing two-sided Cholesky-QR Krylov projections, eliminating SVD stalls, and dynamically steering the projection subspace using router activation geometry.

Matrix Polar and Second-Order Methods: Muon [12] introduced 5th-order Newton-Schulz polynomial iterations to approximate polar decomposition for hidden linear weights. Earlier second-order and preconditioning methods include Shampoo [13,14] and K-FAC [15]. CentroidSteer builds on Muon by restricting polar orthogonalization to 2D attention matrices while establishing specialized low-rank and adaptive update tiers for expert and embedding weights.

Mixture-of-Experts Scaling: Sparse MoE architectures [8–11] drastically expand model capacity while maintaining constant active FLOPs. CentroidSteer represents the first optimizer explicitly co-designed with sparse router activation manifolds.

Table 2: **Multi-Seed 100-Step Convergence on Dense Transformer (43.46M Parameters)** across 3 random seeds (Mean ± Std). All regimes evaluate identical bitwise initial weights.

Optimizer Regime	Final Val Loss	Val Loss Drop	Final Val PPL	Step Time	Peak VRAM	Conve
Standard AdamW (`torch.optim`)	7.6621 ± 0.0577	2.8832 ± 0.0654	2128.7 ± 122.1	32.86 ms	**714.9 MB**	
BitsAndBytes 8-bit AdamW	7.9332 ± 0.0741	2.6121 ± 0.0617	2793.3 ± 202.4	37.07 ms	**632.5 MB**	
Muon Baseline (2D Hidden)	7.5280 ± 0.0061	3.0173 ± 0.0123	1859.3 ± 11.3	**31.00 ms**	863.2 MB	**0**
CentroidSteer (No Muon)	$\mathbf{7.4967 \pm 0.0111}$	$\mathbf{3.0485 \pm 0.0029}$	$\mathbf{1802.2 \pm 19.9}$	33.39 ms	924.8 MB	
Full Triune Stack	7.5280 ± 0.0061	3.0173 ± 0.0123	1859.3 ± 11.3	31.65 ms	863.2 MB	

Table 3: **Multi-Seed Convergence on Mixture-of-Experts (90.79M Parameters)** with 4 routed experts per layer across 3 random seeds. Active Tier 2 parameters: $35,389,440$ (39.0%).

MoE Optimizer Regime	Final Val Loss	Val Loss Drop	Final Val PPL	Step Time	Peak VRAM	Converg
Standard AdamW	7.9216 ± 0.0667	2.5290 ± 0.0623	2760.4 ± 187.6	137.25 ms	**1356.6 MB**	0.
Muon + AdamW Baseline	$\mathbf{7.8246 \pm 0.0041}$	$\mathbf{2.6260 \pm 0.0022}$	$\mathbf{2501.5 \pm 10.4}$	133.23 ms	1364.2 MB	0.
CentroidSteer (No Muon)	7.8860 ± 0.0369	2.5647 ± 0.0413	2660.9 ± 97.8	136.15 ms	1659.7 MB	0.
Triune MoE Stack (3-Tier)	7.8268 ± 0.0558	2.6238 ± 0.0517	2509.5 ± 139.7	**131.71 ms**	1602.0 MB	**0.1**

5 Limitations & Future Work

While CentroidSteer achieves substantial memory savings and solid convergence, several limitations warrant note. First, computing Cholesky-QR basis updates every T_{gap} steps incurs a modest setup latency on the first update step. Second, evaluations in this work were conducted across 100-step pre-training trajectories at 90.8M parameter scale; extended pre-training over billions of tokens will be required to characterize long-horizon scaling laws. Future work will investigate dynamic rank scheduling and distributed tensor-parallel basis synchronization.

6 Conclusion

CentroidSteer demonstrates that low-rank gradient projection and polar orthogonalization can be unified into a cohesive, mathematically grounded optimizer architecture. By guiding expert parameter updates along dynamic router activation manifolds and preserving sensitive embeddings, CentroidSteer reduces persistent expert optimizer memory by 98.4% while providing empirical convergence improvements on modern sparse Mixture-of-Experts models.

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Table 4: **Controlled Routing Steering Ablation on MoE Model (3 Seeds)** under identical initial weights and sequence ordering. Per-seed trajectories and paired statistical tests.

Steering Regime	**Seed 42**	**Seed 1337**	**Seed 2026**	**Mean ± Std**	**Val Loss Drop**	**Paired Δ vs. Unst**
Unsteered Subspace ($\alpha = 0$)	7.7835	7.8767	7.7660	7.8087 ± 0.0595	2.6419 ± 0.0621	—
Random Orthogonal Direction	7.7817	7.7751	7.8616	7.8061 ± 0.0481	2.6445 ± 0.0437	-0.0026 ± 0.098 (p =
Router Centroid Steering	**7.7043**	7.8768	7.8145	$\mathbf{7.7985 \pm 0.0874}$	$\mathbf{2.6521 \pm 0.0863}$	$\mathbf{-0.0102 \pm 0.064}$ (p